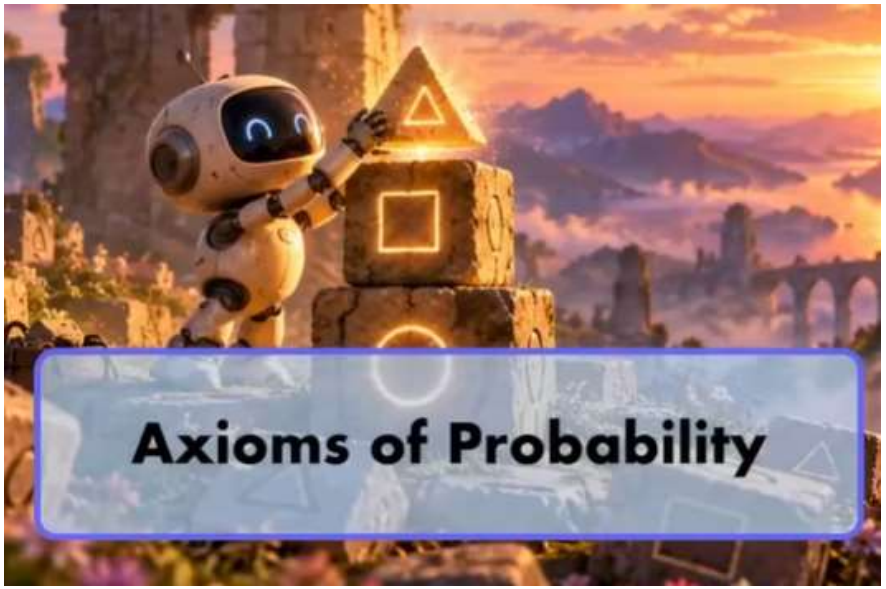
Axioms2

https://youtu.be/Q3ar5nFVuiQ?si=EnyUGOiHAOWqBMgq



A weather prediction algorithm predicts a 0.7 probability that it will rain tomorrow. What is the probability that it doesn't rain, according to the algorithm?

Answer: The probability that it doesn't rain is $1-0.7 = 0.31-0.7 = 0.3$, or **30%**, using the complement rule.

The weather prediction algorithm has a 0.6 probability that it rains at 8am and a 0.5 probability that it rains at 9am. You care whether it rains at either 8am or 9am. Why can't you just add the probability of rain at 8am to the probability of rain at 9am? What adjustment would you need to make to get the probability of 8am or 9am?

Answer: To get the probability of rain at 8am **or** 9am, add the two probabilities and subtract the probability that it rains at both times:

$$\begin{aligned} P(\text{8am or 9am}) &= P(8)+P(9)-P(8 \text{ and } 9) \\ &= 0.6+0.5-P(8 \text{ and } 9) \end{aligned}$$

You would need to know the probability of rain at both times to calculate the exact answer.